

2082:

Long Answer Questions: [10 marks each]

1. List different phases of project management. Explain the different steps involved in the first phase of project management.
2. Compare and contrast between observation and document analysis while requirement determination. Explain radical methods for determining the requirements.
3. Draw DFD for the following scenario:

A patient after registration in the online pathology system will get login credentials. After the valid login the patient upon entering the details of test to be conducted will get the test date and time. The patient can also pay online. The patient can download the test report from the system after notification. The patient will also get schedule for doctor visit if required and in that case the report will also be emailed to the doctor.

Short Answer Questions: [5 marks each]

4. Highlight the critical factors that distinguish agile methods from traditional methods.
5. Illustrate the heart of the software development process.
6. An initial investment of Rs. 5,00,000 is required to start a company. The cash flow-in is Rs. 60,000 per year. Assuming the interest rate as 15%, calculate the discounted payback period. (use Net Present Value)
7. Explain the steps involved in the dialogue design process.

8. Describe the process of designing physical tables.
9. Explain different types of installation.
10. How can maintenance effectiveness be measured? Explain.
11. Draw the sequence diagram for the online food ordering system with the online payment facility.
12. Write short notes on:
 - a. Integration testing
 - b. Relational model

2081:

Long Answer Questions: [10 marks each]

13. Explain prototyping model with a neat diagram. List out its advantages and disadvantages.

14. Draw a reduced decision table for the following scenario:

An applicant has to fill up the form to participate in a music competition. The applicant's age should be greater than 18 years and nationality has to be Nepalese for being selected for the competition. The applicant will be provided a registration number after paying Rs. 5000 as an entry fee. The applicant has to play an instrument for three minutes else will be disqualified. The experts will provide points on the basis of some criteria. The applicant having the highest point will be declared winner and will get a cash prize of Rs. 1,00,000 and all the qualified applicants will get a gift hamper.

15. Differentiate between black box and white box testing. Explain the testing process.

Short Answer Questions: [5 marks each]

16. Briefly explain different phases of project management.

17. How can CASE tool help in developing products? Explain with respect to different phases of SDLC.

18. Draw use case diagram and class diagram for the following scenario:

The admin and user have to log in to the patient record system. The user can see details of its records but will not be able to make changes. The changes will be controlled by admin. The admin has authority to generate reports and send them to doctors and users.

19. How do you think effective interfaces can be designed? Explain.
20. List the advantages of Normalization. Explain 2NF with an example.
21. Explain how disruptive technologies can be used in determining requirements.
22. Difference between process modeling and data modeling. How do you think data modeling plays an important role?
23. An initial investment of Rs. 1,00,000 is required to start a company. The cash flow-in is Rs. 40,000 per year. Assuming interest rate as 10%, calculate the discounted payback period. (use Net Present Value).
24. Write short notes on:
 - c. PERT
 - d. Deliverables

2080:

Long Answer Questions: [10 marks each]

1. What activities are at the heart of systems development process? List and explain some of the problems with the traditional waterfall SDLC. Explain Agile development in brief.
2. What do you mean by balancing DFDs? Draw context diagram and data flow diagrams for online movie rental system that allows its users to watch movies online.
3. Explain physical database design. Why is physical database design important?

Differentiate logical database design with physical database design.

Short Answer Questions: [5 marks each]

4. Describe different activities performed by the project manager during project.
5. Describe the steps involved in corporate strategic planning.
6. Define feasibility study. Explain economic and schedule feasibility in brief.
7. List traditional methods for determining system requirements. Explain advantages and pitfalls of observing workers to determine system requirements.
8. Draw ER diagram to store data about students, programs, and courses at your college.
9. Describe several methods for interacting with the system.
10. What are the major activities of system maintenance? Explain.
11. Explain class diagram with suitable example.
12. Write short notes on:

- a. Network Diagram
- b. Structural and Behavioral Diagrams

Model:

Long Answer Questions: [10 marks each]

1. Explain the modern approach to system analysis and design.
2. Draw a DFD diagram of Mess Management System up to level 2.
3. A bank has the following policy on deposit. On same deposit for a period of Rs. 50,000 and above for five years or above the interest rate is 15%. On same deposit for a period less than 5 year it is 12%. On deposit below Rs.50,000 the interest is 10% regardless of period of deposit. Write the above process using:
 - a. Structured English
 - b. A Decision Table

Short Answer Questions: [5 marks each]

4. Describe five methods of interacting with a system in brief.
5. Define information system. What are the typical components of information systems?
6. Difference between logical database design and physical database design.
7. What do you mean by system implementation? Discuss the steps for system implementation.
8. Who is system analyst? List and explain the skill of system analyst.

9. A system costs Rs. 2,00,000 to install and Rs. 10,000 per month as recurring expenses.

The benefit per year is 150,000. Assuming the interest rate is 15%, what is the payback period of the investment?

10. What are the steps involved in corporate strategic planning?

11. Define feasibility study. Explain different categories of feasibility.

12. What is Use Case Modeling? Explain with example.

2079:

Long Answer Questions: [10 marks each]

1. Differentiate between black box and white box testing. Explain different stages of testing.
2. Why do we use DFD? Draw context diagram and data flow diagrams of a retail clothing store in a mall that sells different clothes to its customers.
3. Compare structured development with object-oriented development. Explain use case diagram and class diagram with suitable example.

Short Answer Questions: [5 marks each]

4. Explain common skills of a project manager. Which skill do you think is most important?
5. Describe the project identification and selection process.
6. List and describe the steps in the project initiation and planning process.
7. Explain Joint Application Design in brief. How is it better than traditional information gathering techniques?
8. What is conceptual data modeling? Explain conceptual data modeling process.
9. Compare form with report. Explain the process of designing forms and reports.
10. Define system maintenance. Explain the process of maintaining information systems in brief.
11. Write short notes on:

- a. CASE tools
- b. Baseline project

2078:

Long Answer Questions: [10 marks each]

1. What is system development life cycle (SDLC)? Explain each phase of SDLC in detail.
2. Assume monetary benefits of an information system of \$70,000 the first year and increasing benefits of \$10,000 a year for the next four years, one-time development costs of \$90,000, and recurring costs of \$40,000 per year over the duration of the system's life. The discount rate for the company was 11 percent. Using a five-year time horizon, calculate the net present value of these costs and benefits. Also, calculate the overall return on investment and then present a breakeven analysis. At what point does breakeven occur?
3. What is conceptual data model? How do you gather information for conceptual data modeling? Draw an ER diagram for a retail store in a mall which sells different items to its customers.

Short Answer Questions: [5 marks each]

4. What are system analysis and design? Why is it important for developing information systems?
5. Why do we need CASE tools during system development? What are different components of CASE tool?
6. What is project management? Explain some activities and skills of a project manager.

7. Explain the process initializing and planning information system development project in brief.
8. How can you use prototyping for determining system prototyping requirements?
Compare throwaway with evolutionary prototyping.
9. What is the purpose of database design? Compare logical design with physical design.
10. What is testing? Explain any four different testing techniques.
11. What is class diagram? Explain class diagram with suitable example.
12. Write short notes on:
 - a. Agile development
 - b. Decision tree

2076:

Long Answer Questions: [10 marks each]

1. What is the waterfall model? Explain the prototyping model for developing information systems along with merits and demerits.
2. Define feasibility. Explain different categories of feasibility. How do you measure economic feasibility?
3. Assuming a retail clothing store in a mall, draw a context diagram and a level 0 diagram that represent the selling system at the store.

Short Answer Questions: [5 marks each]

4. Explain the modern approach to system analysis and design.
5. What is rapid application development? Explain.
6. What is project initiation? Explain different activities you will perform during project initiation phase.
7. What is the process of identifying and selecting information system development project in brief?
8. What is a group interview? What are the benefits and drawbacks of group interviews?
9. How do you format forms and reports? Explain general guidelines for formatting forms and reports.
10. List major activities of maintenance. Explain different types of maintenance activities.

11. What is object-oriented development? How is it different from structured development?

12. Write short notes on

- a. Spiral model
- b. Decision table